

Lab Work 1: “Feature Engineering”

General Remarks

- Please work in groups of two or three people!
- **Clearly mark the code sections that were created with the help of generative AI!**
- Apply train/test data split of 80/20 for every task
- Please use the appropriate sklearn libs

1. Report

- Must be **submitted 5 days before the presentation**
- Must contain the results of all tasks (task 1, task 2, task 3 and task 4.1) clearly readable
- Submit the code commented and easy to read

2. Presentation

- Show and execute the submitted code
- Explain the tasks and show their results
- Comment on the results of your findings

Introduction

The overall goal is to investigate and assess different feature engineering techniques, in particular attribute reduction techniques, which can be applied to binary classification problems. Accuracy will be used for the evaluation only.

For the first task one starts with the original data set. The second task deals with deleting certain attributes and then studies its obtained accuracy. Task 3 and 4 deal with some attribute's reduction techniques.

After completion you have learned:

- Using density analysis and studying its impact on accuracy
- Applying PCA and RFE feature engineering techniques
- Working with gridsearch
- Using proper sklearn libs

Data Set

You will use the so-called “Wine Quality Dataset” from Kaggle¹. It is a publicly available data set that contains about 1400 samples of a Portuguese red wine with 11 attributes,

¹ <https://www.kaggle.com/datasets/yasserh/wine-quality-dataset>

i.e. physio-chemical data and a quality ranking. The quality is based on sensory data, i.e. not evaluated by humans, and has a range of scores from 1 to 8.

The target Y of the classification task is the “quality”. To use the “quality” for a binary classification task, the original data set was modified having only two classes. The first class contains the “bad” wines, all samples with a quality ≤ 5 . The second class contains the “good” wines with a quality > 5 . Since in most cases it is easier to work with numbers than with strings, the first group is from now on referred to as quality = 0 and the second group as quality = 1. The modified data is stored in the file “**Wine_Test_01.csv**”. Fig. 1 shows the first 10 rows of the data set with its 11 attributes and in the last column the changed “quality”, the target Y , contains only 0s and 1s.

fixed acidity	volatile acidity	citric acid	residual sugar	chlorides	free sulfur dioxide	total sulfur dioxide	density	pH	sulphates	alcohol	quality
7.4	0.70	0.00	1.9	0.076	11.0	34.0	0.9978	3.51	0.56	9.4	0
7.8	0.88	0.00	2.6	0.098	25.0	67.0	0.9968	3.20	0.68	9.8	0
7.8	0.76	0.04	2.3	0.092	15.0	54.0	0.9970	3.26	0.65	9.8	0
11.2	0.28	0.56	1.9	0.075	17.0	60.0	0.9980	3.16	0.58	9.8	1
7.4	0.70	0.00	1.9	0.076	11.0	34.0	0.9978	3.51	0.56	9.4	0
7.4	0.66	0.00	1.8	0.075	13.0	40.0	0.9978	3.51	0.56	9.4	0
7.9	0.60	0.06	1.6	0.069	15.0	59.0	0.9964	3.30	0.46	9.4	0
7.3	0.65	0.00	1.2	0.065	15.0	21.0	0.9946	3.39	0.47	10.0	1
7.8	0.58	0.02	2.0	0.073	9.0	18.0	0.9968	3.36	0.57	9.5	1
6.7	0.58	0.08	1.8	0.097	15.0	65.0	0.9959	3.28	0.54	9.2	0

Figure 1: Excerpt of the modified data set with the now binary attribute “quality”.

Task 1: Classification with the Unmodified Attributes

Tasks

- Display a histogram for each attribute
- Show **two histograms** for **each attribute** with respect to Y . Please **comment** on the expected performance of the classifier. Please **explain** how you came to this conclusion.
- Perform **10 runs** of modeling and test

In **each run**, split the dataset **randomly** into training and test (somewhat similar to k-fold cross-validation).

 - For each run a gridsearch with the same sets of hyperparameters, employing the Support Vector Machine (SVM), should be performed.
 - Display the best hyperparameters for each run.
 - With the best hyperparameter for each run, display the obtained Accuracy of the **test data**.
 - Display the average accuracy with its respective standard deviation.

Task 2: Classification after record deletion

The goal of this sub task is to delete some data samples / records of the given data set “**Wine_Test_01.csv**” such that a clear distinction for $Y=0$ and $Y=1$ for one attribute pair is

achieved. Following the source data reduction, study the obtained performance of the classifier.

Basic knowledge and approach²

The goal is to delete specific records and look then at the accuracy achieved by the classifiers. To do this, records are deleted where the specific attribute has a certain value range. Thus, a clear separation of the histogram picks values for $Y=0$ and $Y=1$ for this attribute is achieved. In Fig 2 we applied this to the "volatile acidity" attribute. After deleting records that have a certain values range, a separation of the histograms of the two classes is achieved, as shown in Fig 2. We now observe that for the attribute "volatile acidity" the two clouds that are centered differently for $Y=0$ and $Y=1$ and whose histogram picks are now separated (right histogram). Thus, it should be now less difficult to train the classifier correctly and get a better accuracy.

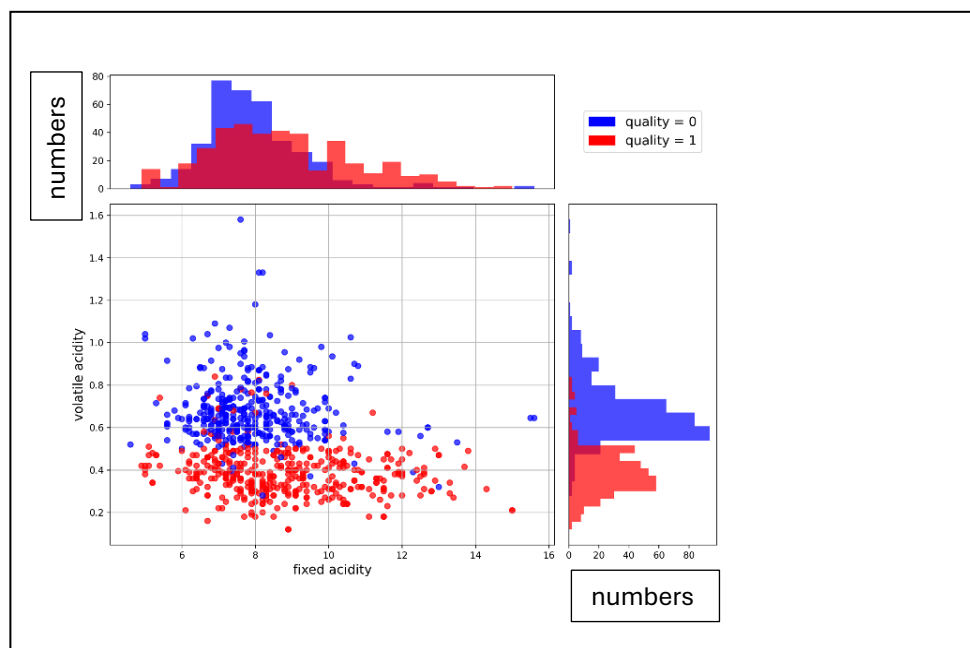


Fig 2: samples deletion gets a separation of the pick for $Y=0$ and $Y=1$

Tasks

- Choose two attributes where there is a significant distribution overlap of $Y=0$ and $Y=1$.
- Perform a density analysis, i.e., do a scatter plot for the chosen attributes and delete samples of a certain range that a separation of the distribution of $Y=0$ and $Y=1$ has been achieved.
- With the modified data source, follow the tasks as listed in **Task 1 c)**.

² jäger-supervised-decision-final-student--2025.pdf, Lecture notes

Task 3: Principal Component Analysis

not required for single performer

The goal for this task is to study the Principal Component Analysis (PCA) technique to be used for a binary classification problem. Apply the PCA to the data from the file “Wine_Test_01.csv”. With the reduced number of principal components, apply then the same task as stated in Task 1 c).

Basic knowledge and approach

PCA is an unsupervised technique for analyzing the importance of attributes (features). Therefore, it can be used to improve a classification algorithm. It provides a useful method for reducing the attribute space when the attributes are highly correlated. The goal of PCA is to eliminate the correlated attributes. The dimensionality is reduced by transforming the attributes into new attributes (features) called Principal Components (PC). The main aim of PCA is to represent the information in a new space requiring less principal components as in the original space^{3,4} for the following modeling. With the obtained PC_i the modeling and test takes place. As the PCA process contains numerically vector/matrix operations, scaling/standardization of the row data set must be done beforehand.

Tasks

The PCA technique should be applied to the **train data** set only. Before After finding the most important attribute by applying the PCA you must make sure that your

- a) Apply PCA on the original data set, using the same number of Principal Components as attributes, and plot “number of components” vs “cumulative variance”, s. Lecture example⁵. Using the proper Classes from the sklearn Libs⁶. From the curve “cumulative variance” vs “components” choose the number of components to be used for the modeling and the test.
- b) With the reduced number of principal components obtained from a) perform the steps given in **Task 1 c)**. For the test make sure that the **test data** are also transformed in the PC space.

³ <https://www.machinelearningplus.com/machine-learning/principal-components-analysis-pca-better-explained/>

⁴ https://en.wikipedia.org/wiki/Principal_component_analysis

⁵ [jaeger-recomender-final-student-2026.pdf](#)

⁶ <https://scikit-learn.org/stable/modules/generated/sklearn.decomposition.PCA.html>

Task 4: Recursive Feature Elimination

not required for single performer

The goal for this sub task is to study and assess one recursive feature elimination (RFE) technique to be used for a binary classification problem. After applying the RFE technique, study the obtained performance of the classifier. Again, the data from the file “Wine_Test_01.csv” should be used.

Basic knowledge and approach

RFE is a technique to analyze the importance of attributes (features) and delete the one which are not so important for the following classification. Thus, it might be a useful task for e.g., reducing the feature space thus saving CPU power.

The RFE technique should be applied to the **train data** set only. After finding the most important attribute by applying the RFE you must make sure that your **test data** have the same attributes as the **training data**.

RFE is an iterative process with the goal of systematically removing the least important attributes the sklearn Libs⁷. It uses the “Y” and therefore the process requires an estimator, i.e., classifier one need to select. A significant difference to the PCA technique where the “Y” is not used. Make sure to use the same estimator chosen for the RFE for the following classification task. Use the proper Classes from the sklearn Libs⁸.

Tasks 4.1

- a) Choose an RFE estimator and with the parameter setting, e.g., “estimator = svm.SVC(kernel='linear' , C = 1000, gamma=0.1)”
- b) Select features, e.g. “selector = RFE(estimator, n_features_to_select=8)”
- c) Apply a RFE and by using the proper classes from sklearn get the names of the attributes. Display them.
- d) With the reduced number of attributes obtained from c) perform modeling with the same estimator chosen for RFE.
- e) With the reduced number of attributes obtained from c) perform the steps given in **Task 1 c)**.

Task 4.2

Perform a RFE technique where the “most important attributes” are being calculated. Try to find a RFE solution where the optimal number of attributes are calculated. Use any help of the net, but you need to cite the source. Perform display the achieved Accuracy and comment on the changes obtained to Task 1c).

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⁷https://en.wikipedia.org/wiki/Feature_selection

⁸ https://scikit-learn.org/stable/modules/generated/sklearn.feature_selection.RFE.html